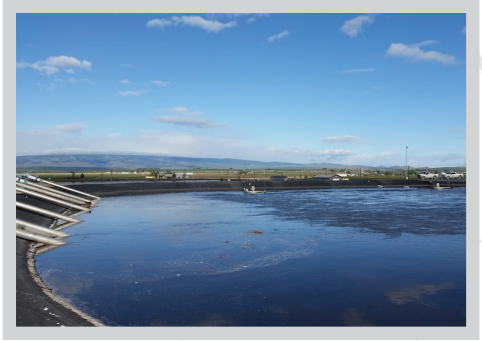
CASE HISTORY:

INDUSTRIAL WWTP - PREDICTIVE AERATION LOGIC (PAL) APPLICATION

CITY OF QUINCY, WA

INSTALL DATE JUNE, 2015



SYNOPSIS

Energy efficient blowers and BioChem's PAL control system were installed at the industrial wastewater treatment plant. These combined technologies are providing a 57% energy savings. This equates to a $97,000 reduction in electric power use per year. The investment including rebates from the Grant County Public Utility District will be fully recovered in just over 3.5 years.

BACKGROUND

The plant treats just over 3MGD using two alternating-duty, sequencing batch reactors (SBR). The project was originally motivated by the failure of one of the plant's 600hp, multi-stage centrifugal blowers. Upon BioChem's recommendation, two 320hp positive displacement blowers with variable frequency drives were installed so that the output of the blowers could be paced to the variable loads associated with the waste stream and SBR process. BioChem provided its patented, self-tuning PAL control software, the main control panel with customized HMI, and integrated the plant's local blower and other control room functions to this panel using Ethernet and Modbus RTU communication as required. BioChem partnered with Pace Engineering and Atlas Copco on this project.

CHALLENGE

This particular project presented various challenges having to do with the physical space to locate the blowers, different communication protocols within the plant, and the need to program an airflow "floor" into the control algorithm in the interest of maintaining minimum mixing even during periods of otherwise low process oxygen demand. These minimum aeration rates for mixing had to be customized for the unique shape and aerator characteristics of each basin. While these challenges were relatively "easy" for BioChem engineers, we like to call attention to this project for the following reasons. Even though SBRs are generally among the least complicated of treatment processes, and even though this plant's cost of energy is less than half the national average, the savings were still dramatic in size.

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EXECUTION

1) BioChem designed and provided the main control panel supporting Modbus, Ethernet IP and analog 4-20mA communication protocols, tying all systems seamlessly together.

2) Installed and customized (minimum mixing requirements) its patented, energy saving PAL control software.

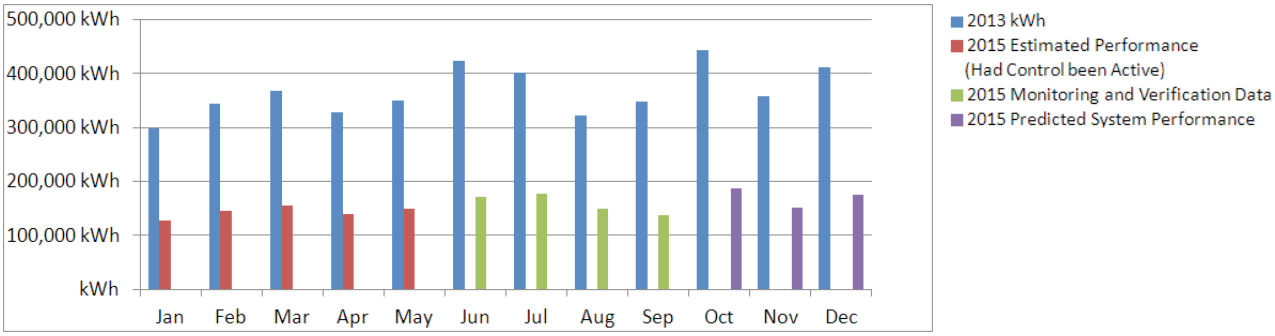
3) Completed full integration and commissioning with less than a week on site.

4) Performance results were monitored over a four month monitoring and verification period against a prior year baseline.

RESULTS

Observed and documented (Pace Engineering) power savings of 57% were achieved with no degradation of process performance. These savings, which translate to $97K annually, and when considering a $240K rebate awarded by the public utility, provide a payback period on the entire project (blowers, VFDs, instruments, software) of just over 3.5 years.

QUINCY ENERGY SAVINGS PROJECTIONS



QUINCY POWER REDUCTION

	2013 power use (kWh)	2015 power use (kWh)	Power reduction (kWh)
May 2015	350,400	205,200	145,200
June 2015	424,200	171,000	253,200
July 2015	401,400	177,600	223,800
August 2015	321,600	148,800	172,800
September 2015	347,400	138,000	209,400
Total (June to September)	**1,494,600**	**635,400**	**859,200 57%**

QUINCY COST SAVINGS

	2013 Power cost	2015 Power Cost	Cost savings
May 2015	$11,211	$6,565	$4,646
June 2015	$11,670	$4,704	$6,966
July 2015	$11,170	$4,942	$6,228
August 2015	$9,554	$4,421	$5,133
September 2015	$8,995	$3,573	$5,422
Sum	**$41,389**	**$17,640**	**$23,749**

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